Michael Liu

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EDUCATION

Carnegie Mellon University Pittsburgh, PA
Bachelor of Science in Computer Science — Concentration in Machine Learning Expected May 2029

- **Selected Coursework:** Computer Systems (15-213), Great Theoretical Ideas in CS (15-251), Functional Programming (15-150), Imperative Computation (15-122), Mathematical Foundations for CS (15-151), Matrices & Linear Transformations (21-241), Probability Theory (36-225), Concepts of Robotics (16-180).

SKILLS

- **Languages:** Python, C, C++, TypeScript, JavaScript, SQL, Java, SML, Bash, x86-64 Assembly.
- **ML & AI:** PyTorch, TensorFlow, CUDA, Scikit-Learn, NumPy, Pandas, OpenCV, RAG, LLM Agents, MCP.
- **Systems & Backend:** FastAPI, React, React Native, Node.js, SQLite, Firebase, Docker, Git, Linux, GDB.
- **Spoken Languages:** Native fluency in English and Mandarin Chinese.

RESEARCH EXPERIENCE

Carnegie Mellon University, VariAbility Lab Pittsburgh, PA
Undergraduate Research Apprentice (SURA) — Advised by Prof. Andrew Begel May 2026 – August 2026

- Integrated the Model Context Protocol (MCP) into Jupyter server extensions to support local context awareness, streamlining developer workflows and reducing context-switching friction.
- Maintained sub-100ms interface responsiveness by engineering custom JupyterLab UI components delivering real-time, non-intrusive emotional support tailored for neurodivergent developers.
- Validated interaction flow and system efficacy by designing and conducting structured qualitative user studies with autistic software engineers, synthesizing key usability insights.

PROJECTS

MarginGuard: AI Financial Watchdog (React, FastAPI, SQLite) | GitHub November 2026

- Cut manual financial reconciliation time by 90%+ by building an asynchronous FastAPI and SQLite ingestion pipeline that extracts and cross-verifies unstructured PDF supplier invoices and POS receipts in under 2s.
- Enabled real-time dish margin tracking and 8-week demand forecasting across 100+ daily POS transactions by integrating weekday-baseline statistical models with a SQL-grounded LLM query agent via OpenRouter.

Viditas: Multimodal Deepfake Detection System August 2026

- Accelerated deepfake classification throughput to 32+ FPS (<31ms/frame) by developing an asynchronous FastAPI and PyTorch backend utilizing CUDA stream concurrency and pinned memory for parallel ingestion.
- Improved multimodal anomaly detection F1-score by 15%+ across benchmark datasets while maintaining <10ms audio-video alignment via a custom presentation-timestamp (PTS) ring buffer.

ScottyTasks (React Native & Firebase) January 2026 – Present

- Boosted recurring daily task completion by 35%+ among CMU student testers by engineering 5+ gamified habit-tracking workflows and interactive progress modules in React Native and Expo.
- Decreased data sync latency to under 50ms across 100% of concurrent sessions by implementing real-time NoSQL state synchronization and user authentication via Google Firebase.

LEADERSHIP & HONORS

CMU ScottyLab (Labrador) Pittsburgh, PA
Project Team Lead & Developer September 2025 – Present

- Guide weekly technical discussions and architectural planning, facilitating cross-functional team ideation while translating high-level feature concepts into concrete implementation milestones.
- Lead sprint planning and adapt project roadmaps in response to technical blockers and shifting scope, keeping teammates aligned on deliverables, code standards, and timeline milestones.

- **CMU NOVA AI Hackathon:** Top 5 Team among People's Choice (November 2026) – DoorDash Back of House Track (*MarginGuard*).
- **Tepper Hackathon (CMU Energy Week):** 3rd Place (March 2026) – Developed a quantitative optimization model analyzing enterprise AI energy demand to improve operational sustainability.
- **USACO Platinum Division (2024):** Perfect score in Gold Division (Top ∼300 nationally).
- **USAMO Qualifier (2024):** AMC 12 top 1% nationwide; scored 11/15 on AIME.